

Problem definition

- Memes are one of the most widely used forms of online communication. Some of them harm individuals and groups by conveying discrimination through a multimodal format that combines image and text.
- How can we mitigate a hateful meme so that the hateful content is removed while preserving the meme's original meaning, humor, and structure?

Related works

- **UnHateMeme [2]:** Suggest a **VLM pipeline** to detect and mitigate multimodal memes; **we extend** it with a **Stable Diffusion model** and **shift to open-source approach**.
- **MemeDetoxNet [3]:** **Localize** and **blurring** sensitive visual regions in the image to reduce harmful content. **We replace global blurring with targeted diffusion-based inpainting** that preserves the surrounding image.

Methods

- Our solution consists of two stages: **Detection** and **Mitigation**.
- In **Detection**, a vision-language model identifies hateful memes and returns a label, a probability and a description.
- In **Mitigation**, the vision-language model localizes the hate source (visual, text, or both), generates an image-editing prompt for Stable Diffusion and suggests new safe text when necessary to produce a mitigated meme.

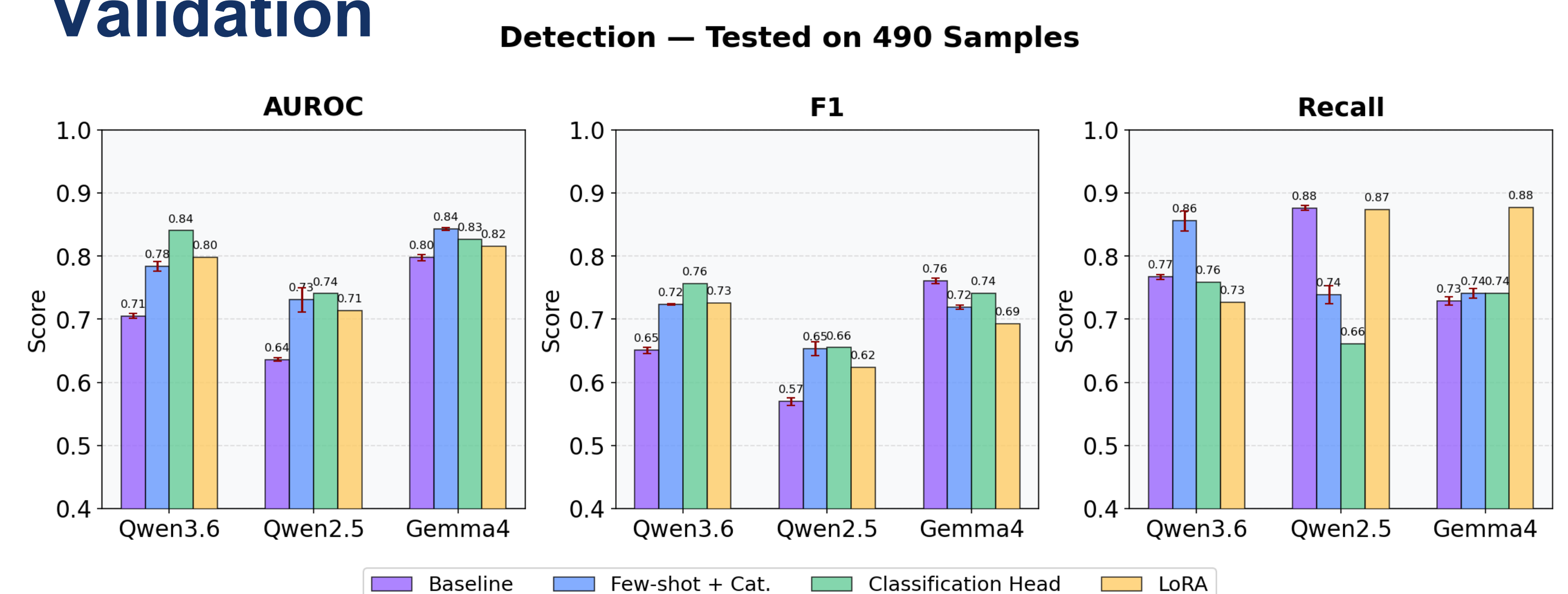


Hateful Meme

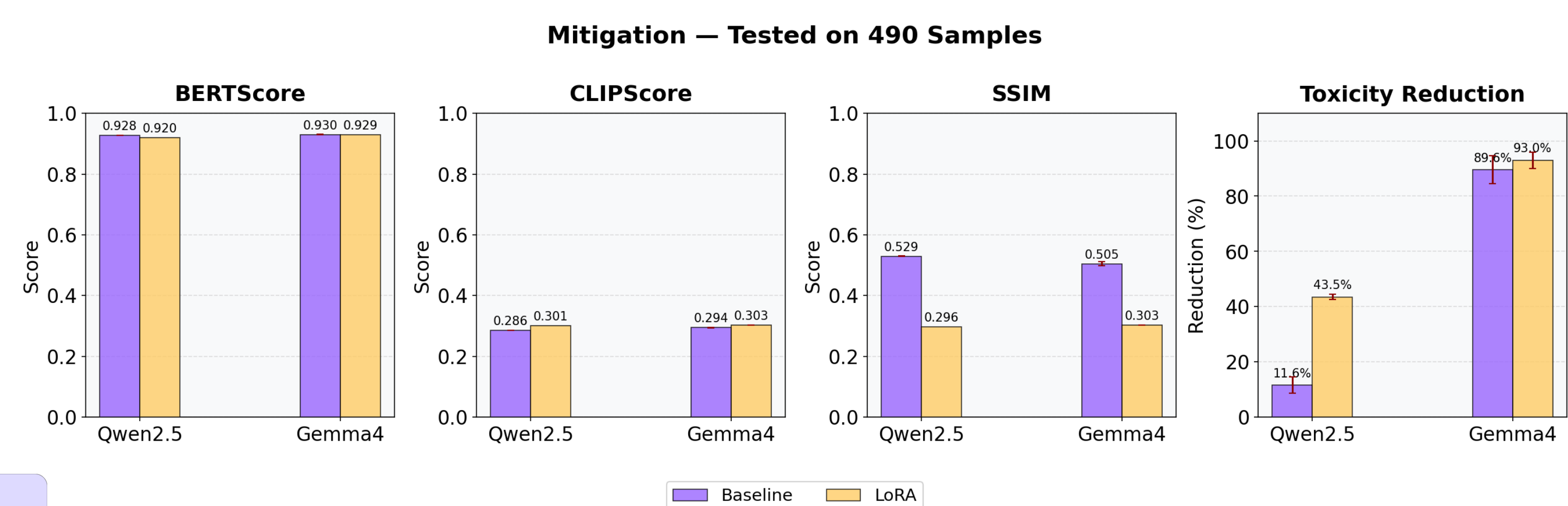


Mitigated Meme

Validation



- **Gemma4-31B is the strongest** (F1 & Recall) but gains least from experiments
- **Few-shot and Classification Head** reliably **beat zero-shot on smaller models** but show more **mixed results on larger models**
- **LoRA Finetuning boosts recall** but **not always AUROC/F1**, it shifts the threshold more than it improves discrimination



- **Gemma4 vastly outperforms Qwen2.5 on toxicity reduction** but have similar results for the other scores
- **BERTScore** (textual sim.) **stays high**, text content is well preserved
- **CLIPScore** (image-text coherence of the mitigated meme) **is low**
- **LoRA halves SSIM** (visual fidelity to the original image) **stronger mitigation** comes at the **cost of visual fidelity**
- **Stable diffusion model used:** black-forest-labs/FLUX.2-klein-9B

Limitations

- **Limited evaluation set** (490 samples)
- **Meaning preservation is hard to measure**; automatic metrics are only proxies
- **Lightweight models**, especially for diffusion
- **Difficult trade-off** between mitigation and meaning preservation

Conclusion

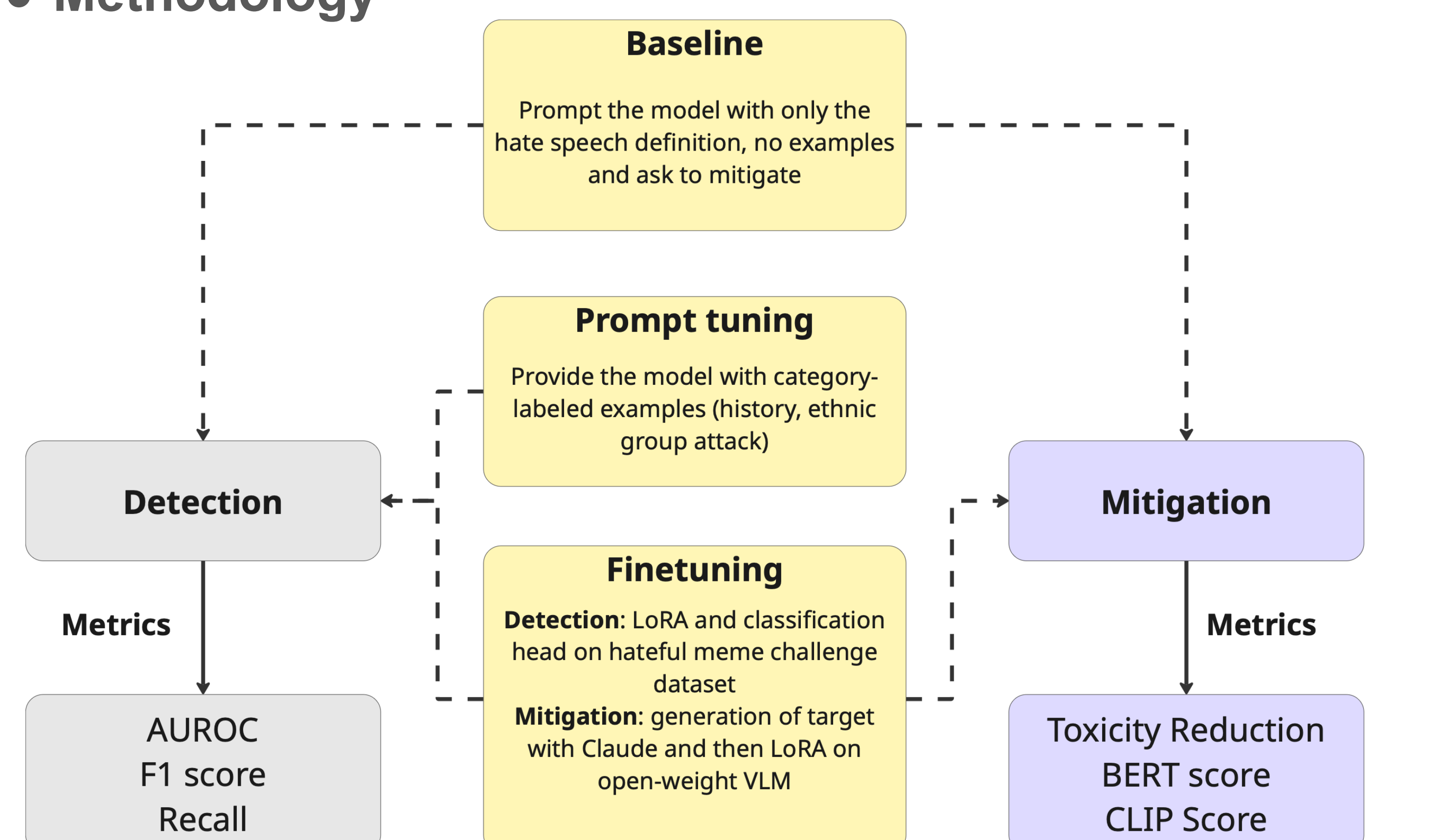
Our pipeline shows **promising results** for detecting and mitigating hate in multimodal memes, demonstrating that a **VLM** acting as both detector and mitigation planner, combined with targeted diffusion-based editing, **can remove hateful content while preserving the original meme's structure and meaning**.

Natural extensions include **scaling beyond memes** and **fine-tuning** the diffusion model on **meme-specific data** to improve the visual quality and coherence.

References

- [1] D. Kiela et al., "The Hateful Memes Challenge: Detecting hate speech in multimodal memes," in Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020, pp. 2611–2624.
- [2] M. M. Van and W. Wu, "Detecting and mitigating hateful content in multimodal memes with vision-language models," arXiv preprint arXiv:2505.00150, 2025.
- [3] G. Kumari, D. Bandyopadhyay, A. Ekbal, V. B. Gangadharaiyah, "MemeDetoxNet: Balancing toxicity reduction and context preservation," in Findings of the Association for Computational Linguistics: ACL 2025, 2025.

Methodology



Datasets

- Facebook Hateful Memes [1]
- 2'000 synthetic data generated with Claude based on Hateful Memes dataset to make finetuning.